

Organizing Scientific Knowledge from the Engineering Sciences Using Research Knowledge Graphs

Dear participant,

We cordially invite you to participate in our survey to evaluate the ongoing work to organize scientific knowledge in engineering using research knowledge graphs. Based on a specific use case on **Tailored Forming Process Chains**, we started to collect, extract, and analyze scientific knowledge from publications. In this context, we have organized various topics to describe process chains in the engineering sciences in a structured manner and thus capture this knowledge systemically.

Your subject matter expertise and feedback are invaluable in assessing the usefulness of the topics and their organization. Furthermore, we would like your opinion on eight competency questions and their answers with visualizations to assess their relevance for Tailored Forming Process Chains in engineering sciences.

The survey is divided into four parts:

1. **Demographics:** Here we ask you to provide some demographic information.
2. **Evaluation of the topics:** Here we ask you to rate the usefulness of the topics for engineering sciences.
3. **Evaluation of the organization of the topics:** Here we ask you to indicate your agreement with the structuring of the topics for engineering sciences.
4. **Evaluation of competency questions:** Here we ask you to evaluate the relevance of competency questions and answers with visualizations for Tailored Forming Process Chains in engineering sciences.

Your answers will be treated confidentially and used exclusively for research purposes.

Participation in the survey takes about **15-20 minutes**.

We thank you in advance for your time and valuable contributions.

Yours sincerely,

Oliver Karras (TIB - Leibniz Information Centre for Science and Technology),
Laura Budde (Laser Zentrum Hannover e.V.),
and Paulina Merkel (IPH - Institut für Integrierte Produktion Hannover gGmbH)

* Gibt eine erforderliche Frage an

Demographics

1. In which subject area of engineering sciences are you a professional? *

We are using the DFG subject area classification. If you need further details to decide if your discipline belongs to one of the subject areas, refer to the following [DFG publication](#).

Engineering Sciences				
Mechanical and Industrial Engineering	Thermal Engineering/ Process Engineering	Materials Science and Engineering	Computer Science, Systems and Electrical Engineering	Construction Engineering and Architecture

Markieren Sie nur ein Oval.

- ☐ Mechanical and Industrial Engineering
- ☐ Thermal Engineering / Process Engineering
- ☐ Material Sciences and Engineering
- ☐ Computer Science, Systems, and Electrical Engineering
- ☐ Construction Engineering and Architecture
- ☐ None

2. What is the highest educational degree you have obtained in engineering sciences? *

3. What is the highest educational degree you have obtained in engineering sciences? *

Markieren Sie nur ein Oval.

- ☐ Bachelor's Degree
- ☐ Master's Degree
- ☐ Doctorate (Ph.D.)
- ☐ Sonstiges: _____

Evaluation of the Topics

Below, we provide definitions of the **topics** for the description of process chains:

- **Materials:** The materials used in a process chain to manufacture a component.
- **Process chains:** The sequences of process steps for manufacturing a component.
- **Process steps:** The steps of a process for manufacturing a component.
- **Inputs:** The inputs, e.g., material or intermediate component, of a process step.
- **Outputs:** The outputs, e.g., intermediate or final component, of a process step.
- **Manufacturing methods:** The manufacturing methods used in a process step to manufacture the output of the process step.
- **Manufactured components:** The output, e.g., intermediate or final component, resulting from a process step.
- **Layers:** The material layers of a manufactured component.
- **Qualities:** The qualities of a manufactured component measured in a process step.
- **Measurement methods:** The measurement methods used to measure the qualities of a manufactured component in a process step
- **Measurements** The measurements, i.e., values and units, of the qualities of a manufactured component in a process step that result from the measurements method used.

4. For each topic listed below, please assess the following statement: The topic ... is **useful** for describing process chains in engineering sciences. *

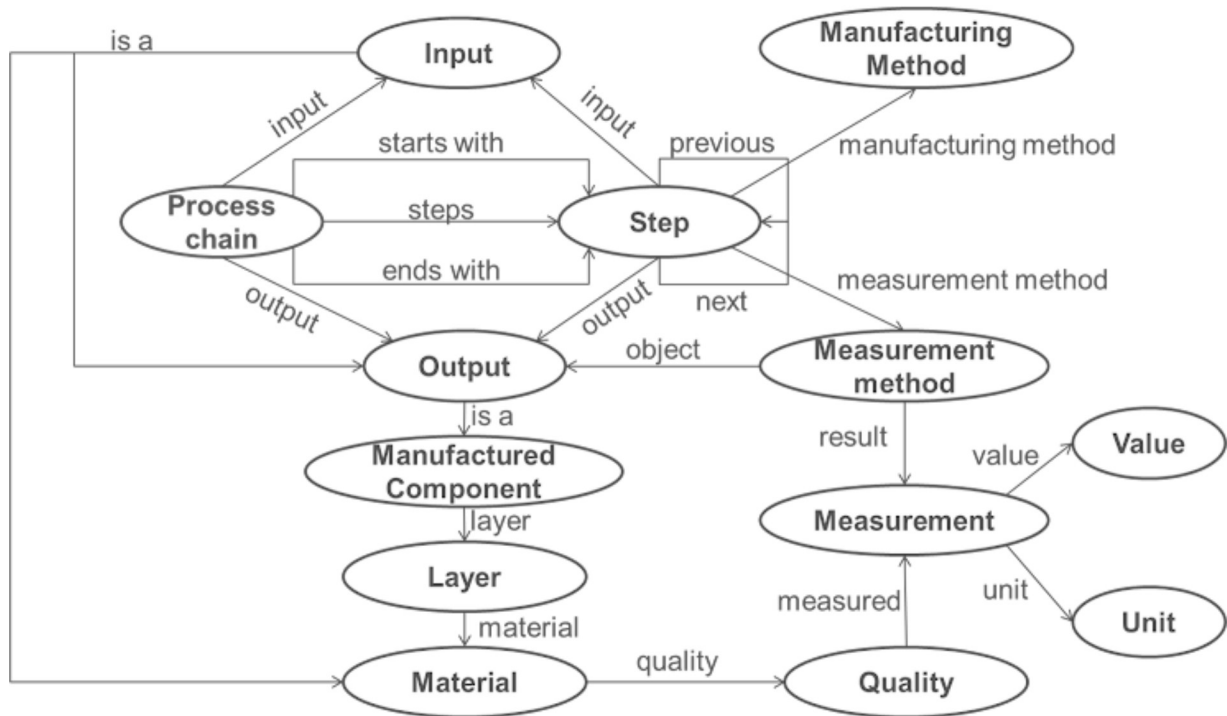
Markieren Sie nur ein Oval pro Zeile.

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
Materials	☐	☐	☐	☐	☐
Process chain	☐	☐	☐	☐	☐
Process steps	☐	☐	☐	☐	☐
Inputs	☐	☐	☐	☐	☐
Outputs	☐	☐	☐	☐	☐
Manufacturing methods	☐	☐	☐	☐	☐
Manufactured components	☐	☐	☐	☐	☐
Layers	☐	☐	☐	☐	☐
Qualities	☐	☐	☐	☐	☐
Measurement methods	☐	☐	☐	☐	☐
Measurements	☐	☐	☐	☐	☐

Evaluation of the Organization of the Topics

Below, we provide an overview of the structuring of the **topics** for the description of process chains.

Structuring of the topics



We have formulated each part of the figure as a statement of our modeling.

Please indicate how much you agree with each of the following ten statements.

5. A **Material** has one or more qualities that are measured. *

Markieren Sie nur ein Oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

6. A **Process chain** has two or more process steps, a start process step, an end ^{*} process step, an input, and an output.

Markieren Sie nur ein Oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly agree

7. A **Process Step** can have a previous process step and a next process step, ^{*} and it has an input, an output, one or more manufacturing methods, and one or more measurement methods.

Markieren Sie nur ein Oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

8. An **Input** is either one or more materials or the output of the previous process ^{*} step.

Markieren Sie nur ein Oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

9. An **Output** is a manufactured component. ^{*}

Markieren Sie nur ein Oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

10. A **Manufacturing method** is the method used in a process step to manufacture the output of the process step. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Agree

11. A **Manufactured component** has one or more layers. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Agree

12. A **Layer** consists of a material. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Agree

13. A **Quality** is measured by one or more measurements. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Agree

14. A **Measurement method** has an output of a process step as the object of investigation and one or more measurements. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Agree

15. A **Measurement** has a value and a unit. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Agree

Evaluation of competency questions

As mentioned earlier, our ongoing work is based on a specific use case on **Tailored Forming Process Chains**, and we started to collect, extract, and analyze scientific knowledge from five publications on ten different Tailored Forming Process Chains. Even though our use case is only a first step, our initial analysis is intended to demonstrate how the knowledge from different publications can be jointly (re-)used to gain new overviews and insights. In particular, we answer eight competency questions asked by two domain experts from engineering sciences based on the organized knowledge.

In the following, we present each competency question and its corresponding answer with a visualization.

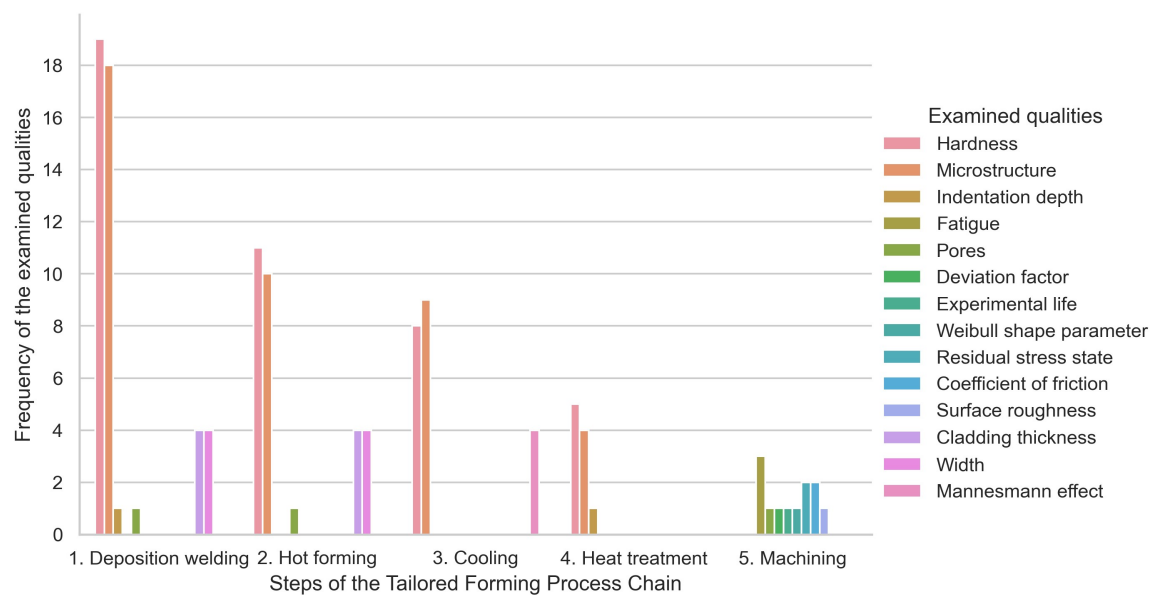
We know that 5 publications and 10 Tailored Forming Process Chains do not allow for generalized results. While our example is small, our basic approach can be scaled to any number of publications.

Considering this scalability, please assess the relevance of each competency question and its answer with the visualization for Tailored Forming Process Chains in engineering sciences.

Competency question 1: How often are certain qualities of hybrid components examined in the individual steps of the Tailored Forming Process Chain?

Based on this question, we want to know which qualities of the manufactured components are typically reported after certain steps of the Tailored Forming Process Chain. On the one hand, this information shows which qualities are reported prominently in scientific publications on Tailored Forming Process Chains. On the other hand, the answer to this question also shows in which steps little research has been reported in general and for specific qualities.

Frequency of the examined qualities per step of the process chain



Answer to Competency Question 1

The figure shows the frequency of the examined qualities in all ten described TFPCs per step. In total, the ten TFPCs report 14 different qualities. The two qualities hardness and microstructure are examined most frequently in four out of five steps. The machining step is the only step where no hardness measurements and microstructure analyses are reported, as no relevant changes in these qualities are expected in this step. Furthermore, we observe that several qualities, such as Weibull shape parameter, experimental life, coefficient of friction, etc., are only measured in the machining step. These qualities are related to the evaluation of the characteristics of the bearing running surface, which can only be examined after the last step, as this step leads to the final running surface of the rolling elements.

Remark: The data for this answer comes from all ten Tailored Forming Process Chains of all five publications.

16. Please assess the following statement: The ... is **relevant** for Tailored Forming Process Chains in engineering sciences.

*

Markieren Sie nur ein Oval pro Zeile.

	Strongly disagree	Disagree	Neutral	Agree	Strongly Agree
Competency question	☐	☐	☐	☐	☐
Answer with visualization	☐	☐	☐	☐	☐

Competency question 2: Which material combinations are used for the manufacturing of hybrid components?

Due to the large number of material combinations for the manufacturing of components, it is necessary to know which material combinations have already been investigated for particular components. On the one hand, this overview allows researchers to find out whether similar studies already exist so that they can compare these results with their results. On the other hand, with this overview, it is possible to find out which material combinations for certain components have not yet been studied and thus represent a gap in science that needs to be closed.

Material combinations used for the manufacturing components

Contribution URL	Layer Component	Material	
		Base material	Cladding layer
http://orkg.org/orkg/resource/R145728	Bearing washer	C22.8	100Cr6
http://orkg.org/orkg/resource/R145731	Bearing washer	C22.8	41Cr4
http://orkg.org/orkg/resource/R145734	Bearing washer	S235	X45CrSi9-3
http://orkg.org/orkg/resource/R162733	Shaft with bearing seat	C22.8	Delcrome 253
http://orkg.org/orkg/resource/R162788	Shaft with bearing seat	C22.8	Stellite 6
http://orkg.org/orkg/resource/R162790	Shaft with bearing seat	C22.8	100Cr6
http://orkg.org/orkg/resource/R171849	Shaft with bearing seat	20MnCr5	X45CrSi9-3
http://orkg.org/orkg/resource/R172160	Shaft with bearing seat	20MnCr5	X45CrSi9-3
http://orkg.org/orkg/resource/R172247	Shaft with bearing seat	20MnCr5	X45CrSi9-3
http://orkg.org/orkg/resource/R172322	Shaft with bearing seat	20MnCr5	X45CrSi9-3

Answer to Competency Question 2

The table shows that a total of three different material combinations were tested for the bearing washer and four different combinations for the shaft with bearing seat. C22.8 is a mild steel and was most frequently used as the base material. However, four variants were also manufactured with 20MnCr5 case-hardening steel as the base material. X45CrSi9-3 and 100Cr6 were used as cladding material for both components, while 41Cr4 or Delcrome and Stellite 6 were only explicitly investigated for one component. Further research potential can be derived from these observations. On the one hand, the influence of the different component geometry and the different process chains on components of the same material combination can be investigated, for example by comparing the different papers. On the other hand, the material spectrum of a component can be expanded to include material combinations that have not yet been investigated, such as C22.8 and Stellite 6 for the bearing washer.

Remark: The data for this answer comes from 10 TFPCs of 5 publications

17. Please assess the following statement: The ... is **relevant** for Tailored Forming Process Chains in engineering sciences.

★

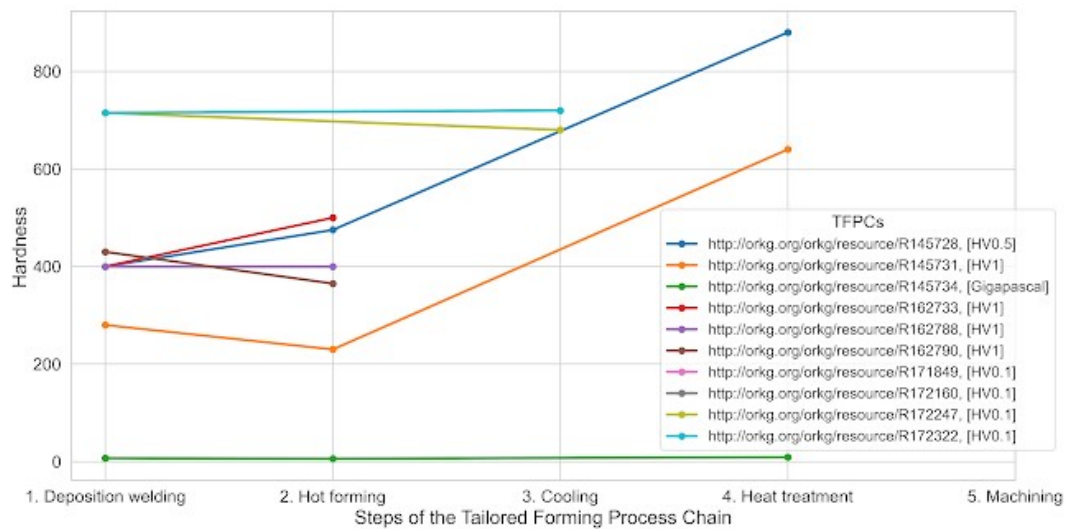
Markieren Sie nur ein Oval pro Zeile.

	Strongly disagree	Disagree	Neutral	Agree	Strongly Agree
Competency question	☐	☐	☐	☐	☐
Answer with visualization	☐	☐	☐	☐	☐

Competency question 3: How do the steps of the Tailored Forming Process Chain affect the hardness of the cladding layer of hybrid components?

The hardness of the cladding layer is an important measure for assessing the suitability of the cladding for mechanically or tribologically highly stressed component areas. For this reason, the question arises as to how the individual steps of the Tailored Forming Process Chain affect the hardness in the cladding layer of a component.

Hardness of the cladding layer of components along the process chain



Answer to Competency Question 3

The figure shows how the hardness of the cladding layer changes in the steps due to the manufacturing methods used. While the publications report the individual values in various places, none of the publications contain a diagram of the change in the hardness along the TFPC steps similar to the one we created. This figure makes it possible to compare the different TFPCs and see how they affect the hardness. Based on this figure, we observe that none of the publications reports the hardness in all steps of the TFPC. While all publications report the hardness in the deposition welding step and six publications in the hot forming step, only four respectively three publications report the hardness in the cooling respectively heat treatment step. No publication reports the hardness after the machining step, as machining does not influence the hardness. In the selected publications, the authors usually report the hardness in the unit Vickers Pyramid Number (HV). Only in one publication, the authors used the unit Gigapascal (GPa). The heat treatment led to an increase in the hardness in all reported cases. The influence of hot forming on hardness depends on the cladding layer material and the specific manufacturing method used for hot forming. Hot forming can lead to an increase or a decrease in hardness compared to the state after deposition welding.

Remark: The data for this answer comes from 10 TFPCs of 5 publications.

18. Please assess the following statement: The ... is **relevant** for Tailored Forming Process Chains in engineering sciences.

*

Markieren Sie nur ein Oval pro Zeile.

	Strongly disagree	Disagree	Neutral	Agree	Strongly Agree
Competency question	☐	☐	☐	☐	☐
Answer with visualization	☐	☐	☐	☐	☐

Competency question 4: In which material combinations and steps of the Tailored Forming Process Chain can defects occur in hybrid components?

Various defects, such as pores and cracks, can occur during the manufacturing of hybrid components. It is important to know at which steps and material combinations these defects occur to assess the applicability of the Tailored Forming process chain for component manufacturing and to identify optimization potential.

Defects in martial combinations and steps

	Contribution URL	Component	Layer with Defect	Material with Defect	Other Layer	Other Material	Defect	Step Name	Step Number
0	http://orkg.org/orkg/resource/R145728	Bearing washer	Cladding layer	100Cr6	Base material	C22.8	Pores	Machining	5
9	http://orkg.org/orkg/resource/R145731	Bearing washer	Cladding layer	41Cr4	Base material	C22.8	Pores	Deposition welding	1
15	http://orkg.org/orkg/resource/R145731	Bearing washer	Cladding layer	41Cr4	Base material	C22.8	Pores	Hot forming	2
24	http://orkg.org/orkg/resource/R171649	Shaft with bearing seat	Base material	20MnCr5	Cladding layer	X45CrSi9-3	Mannesmann effect	Cooling	3
30	http://orkg.org/orkg/resource/R172160	Shaft with bearing seat	Base material	20MnCr5	Cladding layer	X45CrSi9-3	Mannesmann effect	Cooling	3
37	http://orkg.org/orkg/resource/R172247	Shaft with bearing seat	Base material	20MnCr5	Cladding layer	X45CrSi9-3	Mannesmann effect	Cooling	3
47	http://orkg.org/orkg/resource/R172322	Shaft with bearing seat	Base material	20MnCr5	Cladding layer	X45CrSi9-3	Mannesmann effect	Cooling	3

Answer to Competency Question 4

The table shows various defects detected at different steps in the process chain. The Mannesmann effect occurred after cooling following cross-wedge rolling when the material combination 20MnCr5/X45CrSi9-3 was used. With other material combinations, no defects occurred during the production of the shaft. Pores occurred in the bearing washer. These were detected after deposition welding (41Cr4 cladding) and were still present after hot forming. With the 100Cr6 cladding, the pores were detected after machining. The suitability of the material combinations for the Tailored Forming Process Chain can be assessed based on the overview. The processes responsible for the formation of defects can be identified.

Remark: The data for this answer comes from 6 TFPCs of 3 publications.

19. Please assess the following statement: The ... is **relevant** for Tailored Forming Process Chains in engineering sciences.

*

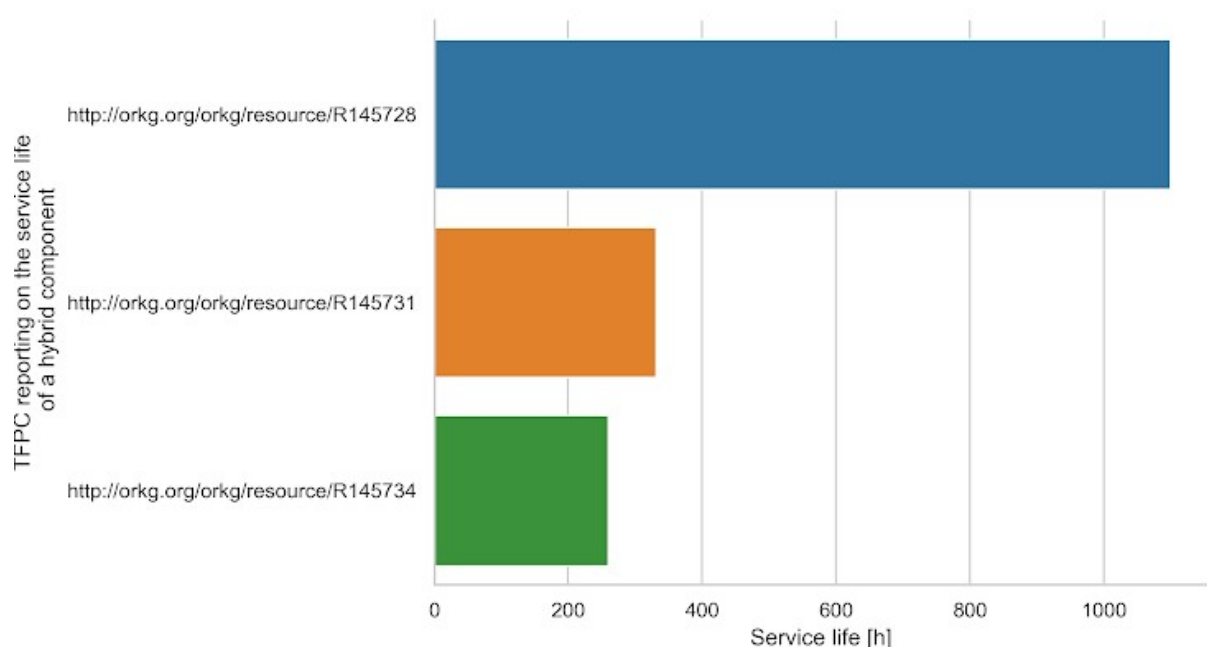
Markieren Sie nur ein Oval pro Zeile.

	Strongly disagree	Disagree	Neutral	Agree	Strongly Agree
Competency question	☐	☐	☐	☐	☐
Answer with visualization	☐	☐	☐	☐	☐

Competency question 5: What service lives can be achieved for hybrid components in the Tailored Forming Process Chain?

An important criterion for components subjected to rolling loads is the service life, which is determined on test rigs. The service life depends on the materials used and other parameters such as hardness and number of defects.

Service life of manufactured components



Answer to Competency Question 5

The figure presents the reported service lives of different hybrid components. In total, only three out of the ten TFPCs report a service life for the manufactured component, which substantiates that this information and question are specific. We observe that there is one TFPC that has led to more than three times higher service life for the manufactured hybrid component than the other two TFPCs. This finding highlights the usefulness of this type of analysis, as it helps not only to compare TFPCs with each other but also to find TFPCs with specific results, such as a manufactured component with a long service life. Based on this finding, researchers can take a closer look at the associated publication to better understand how this TFPC could lead to this long service life.

Remark: The data for this answer comes from 3 TFPCs of 3 publications.

20. Please assess the following statement: The ... is **relevant** for Tailored Forming Process Chains in engineering sciences.

*

Markieren Sie nur ein Oval pro Zeile.

	Strongly disagree	Disagree	Neutral	Agree	Strongly Agree
Competency question	☐	☐	☐	☐	☐
Answer with visualization	☐	☐	☐	☐	☐

Competency question 6: Which manufacturing methods are used in the individual steps of the Tailored Forming Process Chain to manufacture the respective hybrid component?

In addition to the higher-level steps of the Tailored Forming Process Chain, e.g., Hot Forming, it is also of interest to know exactly which manufacturing methods were used in the individual steps to manufacture the hybrid components. This overview therefore shows in detail the manufacturing methods used in each step and in which order.

Service life of manufactured components

	Step number	1	2	3	4	5
	Step name	Deposition welding	Hot forming	Cooling	Heat treatment	Machining
Tailored Forming Process Chain (TFPC)	Hybrid component					
http://orkg.org/orkg/resource/R145728	Bearing washer	Plasma-transferred arc welding	Upsetting	--	Hardening	Grinding
http://orkg.org/orkg/resource/R145731	Bearing washer	Plasma-transferred arc welding	Die forging	--	Hardening	Turning
http://orkg.org/orkg/resource/R145734	Bearing washer	Laser hot-wire cladding	Die forging	--	Hardening	Turning
http://orkg.org/orkg/resource/R162733	Shaft with bearing seat	Plasma-transferred arc welding	Cross-wedge-rolling	--	--	--
http://orkg.org/orkg/resource/R162788	Shaft with bearing seat	Plasma-transferred arc welding	Cross-wedge-rolling	--	--	--
http://orkg.org/orkg/resource/R162790	Shaft with bearing seat	Plasma-transferred arc welding	Cross-wedge-rolling	--	--	--
http://orkg.org/orkg/resource/R171849	Shaft with bearing seat	Laser hot-wire cladding	Cross-wedge-rolling	Air flow	--	--
http://orkg.org/orkg/resource/R172160	Shaft with bearing seat	Laser hot-wire cladding	Cross-wedge-rolling	Water bath	--	--
http://orkg.org/orkg/resource/R172247	Shaft with bearing seat	Laser hot-wire cladding	Cross-wedge-rolling	Air flow	--	--
http://orkg.org/orkg/resource/R172322	Shaft with bearing seat	Laser hot-wire cladding	Cross-wedge-rolling	Water bath	--	--

Answer to Competency Question 6

For each of the ten TFPCs, the figure shows the hybrid component and the manufacturing methods used in each step of the respective TFPC. While all publications report some details on the manufacturing methods used, none of them provide details on all steps of the TFPC. First, we observe that a manufacturing method is always reported for the deposition welding and hot forming step. In particular, plasma-transferred arc welding and laser hot-wire cladding are each used five times as a manufacturing method for deposition welding. Three different manufacturing methods are used for hot forming, with cross-wedge rolling being used most frequently (seven times). For the subsequent steps of the TFPCs, information about the manufacturing method used is often missing in the associated publications.

After consulting with the domain experts who asked the competency questions, we found that they interpreted the missing data as follows: 1) A missing manufacturing method for the cooling step is interpreted as air cooling, and 2) A missing manufacturing method for the subsequent heat treatment and machining steps are interpreted to mean that the steps were not performed. This interpretation is critical as it can lead to incorrect conclusions. As the data is stored in a research knowledge graph, we follow the open-world assumption, meaning missing information is interpreted as unknown information and not as negative information, such as steps not performed. For this reason, the domain experts' interpretation of the missing data is not well-founded, as there is no evidence in the publications and therefore may lead to incorrect conclusions. This way of interpreting missing data shows how important it is for engineers to document all steps of an engineering (research) process in detail to guarantee the trustworthiness of published results and their reproducibility.

Remark: The data of this answer comes from 10 TFPCs of 5 publications.

21. Please assess the following statement: The ... is **relevant** for Tailored Forming Process Chains in engineering sciences.

*

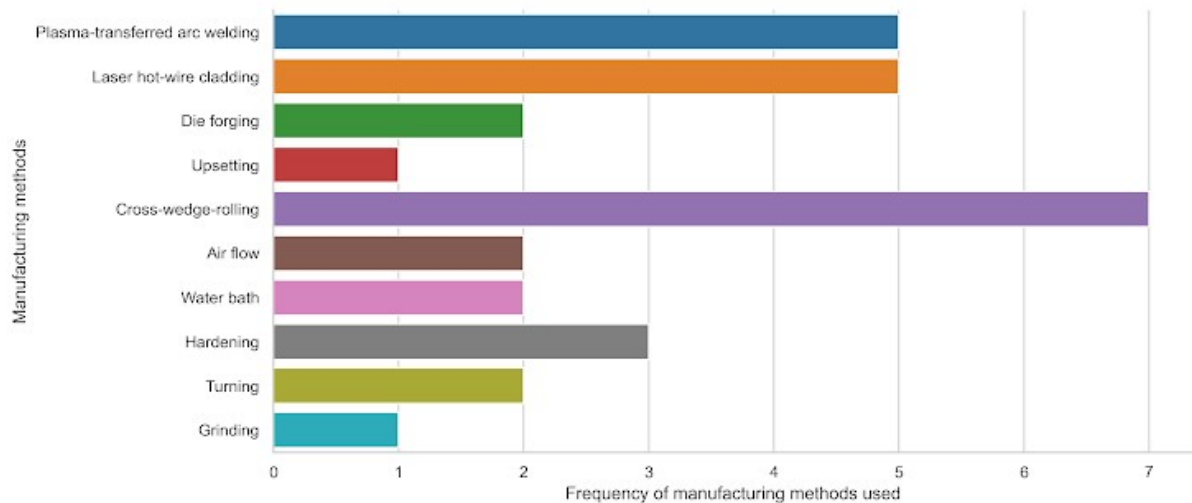
Markieren Sie nur ein Oval pro Zeile.

	Strongly disagree	Disagree	Neutral	Agree	Strongly Agree
Competency question	☐	☐	☐	☐	☐
Answer with visualization	☐	☐	☐	☐	☐

Competency question 7: How often are certain manufacturing methods used in total in all described Tailored Forming Process Chains?

The overview of the manufacturing methods used and their frequency makes it possible to determine previous research priorities and identify research gaps.

Frequency of the manufacturing methods used in total in all described process chains



Answer to Competency Question 7

The figure shows the number of manufacturing methods used for all TFPCs of the 5 publications. In the publications reviewed, cross-wedge rolling was examined most frequently, followed by the two cladding methods laser hot-wire cladding and plasma-transferred arc welding. The fewest findings are available on the later process steps in the Tailored Forming Process Chain, such as grinding and turning. Based on the analysis, it can be determined that in the publications examined, cross-wedge rolling was most frequently used as a forming process and only a few studies on die forging and upsetting are available. In addition, the entire process chain was only completed in a few cases, so there is a need for further investigations.

Remark: The data of this answer comes from 10 TFPCs of 5 publications.

22. Please assess the following statement: The ... is **relevant** for Tailored Forming Process Chains in engineering sciences.

*

Markieren Sie nur ein Oval pro Zeile.

	Strongly disagree	Disagree	Neutral	Agree	Strongly Agree
Competency question	☐	☐	☐	☐	☐
Answer with visualization	☐	☐	☐	☐	☐

Competency question 8: Which microstructures are present in hybrid components after the steps of the Tailored Forming Process Chain?

After each step of the Tailored Forming Process Chain, changes in the microstructure may occur due to the manufacturing method used. Knowledge of these microstructural changes is necessary for the design of a process chain. An overview of the microstructure in the cladding layer and the base material after the steps of the respective Tailored Forming process Chain is given.

Microstructures after the steps of each process chain

	Contribution URL	Component	Layer	Material	Microstructure	Step Name	Step Number
2	http://orkg.org/orkg/resource/R145728	Bearing washer	Base material	C22.8	ferritic-pearlitic	Deposition welding	1
4	http://orkg.org/orkg/resource/R145728	Bearing washer	Cladding layer	100Cr6	pearlitic	Deposition welding	1
7	http://orkg.org/orkg/resource/R145728	Bearing washer	Base material	C22.8	ferritic-pearlitic	Hot forming	2
9	http://orkg.org/orkg/resource/R145728	Bearing washer	Cladding layer	100Cr6	pearlitic	Hot forming	2
12	http://orkg.org/orkg/resource/R145728	Bearing washer	Base material	C22.8	ferritic-pearlitic and martensite	Heat treatment	4
14	http://orkg.org/orkg/resource/R145728	Bearing washer	Cladding layer	100Cr6	martensite	Heat treatment	4
27	http://orkg.org/orkg/resource/R145731	Bearing washer	Base material	C22.8	ferritic-pearlitic	Deposition welding	1
29	http://orkg.org/orkg/resource/R145731	Bearing washer	Cladding layer	41Cr4	pearlitic	Deposition welding	1
33	http://orkg.org/orkg/resource/R145731	Bearing washer	Base material	C22.8	ferritic-pearlitic	Hot forming	2
35	http://orkg.org/orkg/resource/R145731	Bearing washer	Cladding layer	41Cr4	ferritic-pearlitic	Hot forming	2
38	http://orkg.org/orkg/resource/R145731	Bearing washer	Base material	C22.8	martensite	Heat treatment	4
40	http://orkg.org/orkg/resource/R145731	Bearing washer	Cladding layer	41Cr4	martensite	Heat treatment	4
55	http://orkg.org/orkg/resource/R162733	Shaft with bearing seat	Base material	C22.8	ferritic	Deposition welding	1
57	http://orkg.org/orkg/resource/R162733	Shaft with bearing seat	Cladding layer	Delcrome 253	The used etching of 2% nitric acid is not able...	Deposition welding	1
59	http://orkg.org/orkg/resource/R162733	Shaft with bearing seat	Base material	C22.8	ferritic-pearlitic	Hot forming	2
61	http://orkg.org/orkg/resource/R162733	Shaft with bearing seat	Cladding layer	Delcrome 253	The used etching of 2% nitric acid is not able...	Hot forming	2
63	http://orkg.org/orkg/resource/R162788	Shaft with bearing seat	Base material	C22.8	ferritic	Deposition welding	1
65	http://orkg.org/orkg/resource/R162788	Shaft with bearing seat	Cladding layer	Stellite 6	The used etching of 2% nitric acid is not able...	Deposition welding	1
67	http://orkg.org/orkg/resource/R162788	Shaft with bearing seat	Base material	C22.8	ferritic-pearlitic	Hot forming	2
69	http://orkg.org/orkg/resource/R162788	Shaft with bearing seat	Cladding layer	Stellite 6	The used etching of 2% nitric acid is not able...	Hot forming	2
71	http://orkg.org/orkg/resource/R162790	Shaft with bearing seat	Base material	C22.8	ferritic	Deposition welding	1
73	http://orkg.org/orkg/resource/R162790	Shaft with bearing seat	Cladding layer	100Cr6	pearlitic	Deposition welding	1
75	http://orkg.org/orkg/resource/R162790	Shaft with bearing seat	Base material	C22.8	ferritic-pearlitic	Hot forming	2
77	http://orkg.org/orkg/resource/R162790	Shaft with bearing seat	Cladding layer	100Cr6	pearlitic	Hot forming	2
80	http://orkg.org/orkg/resource/R171849	Shaft with bearing seat	Base material	20MnCr5	ferritic-pearlitic	Deposition welding	1
84	http://orkg.org/orkg/resource/R171849	Shaft with bearing seat	Cladding layer	X45CrSi9-3	martensite and austenite	Deposition welding	1
90	http://orkg.org/orkg/resource/R171849	Shaft with bearing seat	Base material	20MnCr5	ferritic-pearlitic	Cooling	3
93	http://orkg.org/orkg/resource/R171849	Shaft with bearing seat	Cladding layer	X45CrSi9-3	martensite	Cooling	3
97	http://orkg.org/orkg/resource/R172160	Shaft with bearing seat	Base material	20MnCr5	ferritic-pearlitic	Deposition welding	1
101	http://orkg.org/orkg/resource/R172160	Shaft with bearing seat	Cladding layer	X45CrSi9-3	martensite and austenite	Deposition welding	1
108	http://orkg.org/orkg/resource/R172160	Shaft with bearing seat	Base material	20MnCr5	martensite	Cooling	3
112	http://orkg.org/orkg/resource/R172160	Shaft with bearing seat	Cladding layer	X45CrSi9-3	martensite	Cooling	3
115	http://orkg.org/orkg/resource/R172247	Shaft with bearing seat	Base material	20MnCr5	ferritic-pearlitic	Deposition welding	1
119	http://orkg.org/orkg/resource/R172247	Shaft with bearing seat	Cladding layer	X45CrSi9-3	martensite and austenite	Deposition welding	1
126	http://orkg.org/orkg/resource/R172247	Shaft with bearing seat	Base material	20MnCr5	ferritic-pearlitic	Cooling	3
132	http://orkg.org/orkg/resource/R172247	Shaft with bearing seat	Cladding layer	X45CrSi9-3	martensite	Cooling	3
133	http://orkg.org/orkg/resource/R172247	Shaft with bearing seat	Cladding layer	X45CrSi9-3	martensite	Cooling	3
137	http://orkg.org/orkg/resource/R172322	Shaft with bearing seat	Base material	20MnCr5	ferritic-pearlitic	Deposition welding	1
141	http://orkg.org/orkg/resource/R172322	Shaft with bearing seat	Cladding layer	X45CrSi9-3	martensite and austenite	Deposition welding	1
147	http://orkg.org/orkg/resource/R172322	Shaft with bearing seat	Base material	20MnCr5	martensite	Cooling	3
152	http://orkg.org/orkg/resource/R172322	Shaft with bearing seat	Cladding layer	X45CrSi9-3	martensite	Cooling	3

Answer to Competency Question 8

The table shows that different microstructures occur in the base material and cladding material. The base material is usually ferritic-pearlitic, but a martensitic microstructure occurs after heat treatment. A similar trend can be observed for the cladding materials 41Cr4 and 100Cr6. For the materials Delcrome 253 and Stellite 6, no microstructure could be identified with the etchant used.

Remark: The data for this answer comes from 9 TFPCs of 4 publications.

23. Please assess the following statement: The ... is **relevant** for Tailored Forming Process Chains in engineering sciences.

*

Markieren Sie nur ein Oval pro Zeile.

	Strongly disagree	Disagree	Neutral	Agree	Strongly Agree
Competency question	☐	☐	☐	☐	☐
Answer with visualization	☐	☐	☐	☐	☐

Thank You For Your Participation

We appreciate you taking the time to complete our survey. Your insights and expertise are invaluable to us and will greatly contribute to the success of our research.

If you have any additional comments or suggestions, please feel free to share them by email. We are always looking for ways to improve and value your feedback.

Help Us Reach More Experts!

If you know other professionals from engineering sciences who might be interested in participating, please share this survey with them. Your support in spreading the word is greatly appreciated!

If you have any questions or need further assistance, please contact us at oliver.karras@tib.eu.

Thank you once again for your valuable contribution!

Best regards,

Oliver Karras (TIB - Leibniz Information Centre for Science and Technology),
Laura Budde (Laser Zentrum Hannover e.V.),
and Paulina Merkel (IPH - Institut für Integrierte Produktion Hannover gGmbH)

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